

### CONCEPT 1.3

#### In studying nature, scientists form and test hypotheses (pp. 16–22)

- In scientific **inquiry**, scientists collect **data** and use **inductive reasoning** to draw a general conclusion, which can be developed into a testable **hypothesis**. **Deductive reasoning** uses predictions to test hypotheses. Hypotheses must be testable; science can address neither the possibility of supernatural phenomena nor religious beliefs. Hypotheses can be tested by conducting **experiments** or, when that is not possible, by making observations. In the process of science, the core activity is testing ideas. This endeavor is influenced by exploration and discovery, community analysis and feedback, and societal outcomes.
- **Controlled experiments** are designed to demonstrate the effect of one **variable** by testing control groups and experimental groups that differ in only that one variable.
- A scientific **theory** is broad in scope, generates new hypotheses, and is supported by a large body of evidence.

? What are the roles of gathering and interpreting data?

### CONCEPT 1.4

#### Science benefits from a cooperative approach and diverse viewpoints (pp. 22–24)

- Science is a social activity. The work of each scientist builds on the work of others who have come before. Scientists must be able to repeat each other's results, and integrity is key. Biologists approach questions at different levels; their approaches complement each other.
- **Technology** consists of any method or device that applies scientific knowledge for some specific purpose that affects society. The impact of basic research is not always immediately obvious.
- Diversity among scientists promotes progress in science.

? Explain why different approaches and diverse backgrounds among scientists are important.

## TEST YOUR UNDERSTANDING

➔ For more multiple-choice questions, go to the **Practice Test** in the **Mastering Biology** eText or Study Area, or go to [goo.gl/GruWRg](http://goo.gl/GruWRg).

### Levels 1-2: Remembering/Understanding

1. All the organisms on your campus make up
  - (A) an ecosystem.
  - (B) a community.
  - (C) a population.
  - (D) a taxonomic domain.
2. Systems biology is mainly an attempt to
  - (A) analyze genomes from different species.
  - (B) simplify complex problems by reducing the system into smaller, less complex units.
  - (C) understand the behavior of entire biological systems by studying interactions among its component parts.
  - (D) build high-throughput machines to rapidly acquire data.
3. Which of these best demonstrates unity among organisms?
  - (A) emergent properties
  - (B) descent with modification
  - (C) the structure and function of DNA
  - (D) natural selection
4. A controlled experiment is one that
  - (A) proceeds slowly so a scientist can make careful records.
  - (B) tests experimental and control groups in parallel.
  - (C) is repeated many times to make sure the results are accurate.
  - (D) keeps all variables constant.

5. Which of the following statements best distinguishes hypotheses from theories in science?
  - (A) Theories are hypotheses that have been proved.
  - (B) Hypotheses are guesses; theories are correct answers.
  - (C) Hypotheses usually are relatively narrow in scope; theories have broad explanatory power.
  - (D) Theories are proved true; hypotheses are often contradicted by experimental results.

### Levels 3-4: Applying/Analyzing

6. Which of the following is an example of qualitative data?
  - (A) The fish swam in a zigzag motion.
  - (B) The contents of the stomach are mixed every 20 seconds.
  - (C) The temperature decreased from 20°C to 15°C.
  - (D) The six pairs of robins hatched an average of three chicks each.
7. Which sentence best describes the logic of scientific inquiry?
  - (A) If I generate a testable hypothesis, tests and observations will support it.
  - (B) If my prediction is correct, it will lead to a testable hypothesis.
  - (C) If my observations are accurate, they will support my hypothesis.
  - (D) If my prediction turns out to be correct, my hypothesis is supported.
8. **DRAW IT** Draw a biological hierarchy similar to the one in Figure 1.3 but using a coral reef as the ecosystem, a fish as the organism, its stomach as the organ, and DNA as the molecule. Include all levels in the hierarchy.

### Levels 5-6: Evaluating/Creating

9. **EVOLUTION CONNECTION** A typical prokaryotic cell has about 3,000 genes in its DNA, while a human cell has about 21,300 genes. About 1,000 of these genes are present in both types of cells. Explain how such different organisms could have this same subset of 1,000 genes. What sorts of functions might these shared genes have?
10. **SCIENTIFIC INQUIRY** Based on the results of the mouse coloration case study, suggest another hypothesis researchers might use to study the role of predators in natural selection.
11. **SCIENTIFIC INQUIRY** Scientists search the scientific literature using electronic databases such as PubMed, a free online database maintained by the National Center for Biotechnology Information. Use PubMed to find the abstract of an article that Hopi Hoekstra published in 2017 or later.
12. **WRITE ABOUT A THEME: EVOLUTION** In a short essay (100–150 words), discuss Darwin's view of how natural selection resulted in both unity and diversity of life. Include in your discussion some of his evidence. (For help in writing essays, see "Writing Tips and Rubrics" in the Study Area of Mastering Biology under "Additional Resources.")
13. **SYNTHESIZE YOUR KNOWLEDGE**



Can you pick out the mossy leaf-tailed gecko lying against the tree trunk in this photo? How is the appearance of the gecko a benefit in terms of survival? Given what you learned about evolution, natural selection, and genetic information in this chapter, describe how the gecko's coloration might have evolved.

For selected answers, see Appendix A.